Eye-Based Cursor Controller using Facial Tracking

Presented by – Group 9

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Problem Statement

Input:

- Live video stream from webcam

Output:

- Cursor movement
- Mouse clicks (left/right)
- Scrolling

Constraints:

- Real-time performance (low latency) (upto 150ms)
- No additional hardware (only webcam)
- Robust to natural blinks and head movement

Motivation

- Enables accessibility for users with limited motor control
- Useful in hands-free environments (labs, presentations)
- Reduces dependency on physical input devices
- Existing solutions are often:
  - Expensive
  - Require specialized hardware
  - Computationally heavy or unstable

Literature Review

What has been done before?

- A simple virtual mouse system was developed using OpenCV, MediaPipe, and PyAutoGUI. The implementation focused on detecting facial landmarks and controlling the cursor through the movement of the right eye. Core functionalities included cursor navigation along with basic left and right click operations.
- An enhanced version of the system introduced additional capabilities such as scrolling and gesture-based controls for performing left and right click actions more intuitively.

Literature Review

How our work is different?

- Our approach extends prior implementations by adapting them to improve usability in presentation and lecture environments.
- A key enhancement lies in the gesture recognition system, which has been broadened to include actions such as scrolling up, scrolling down, and switching between mouse-based and eye-based control modes, making the system more flexible for presentation tasks.

Proposed Method

Cursor Movement

- Track right iris (landmark 477)
- Map normalized coordinates → screen space

Click Detection

- Eye Aspect Ratio (EAR) used
- EAR < 0.20 → eye closed
- Left wink → Left click
- Right wink → Right click
- Debounce: 500 ms (avoid false clicks)

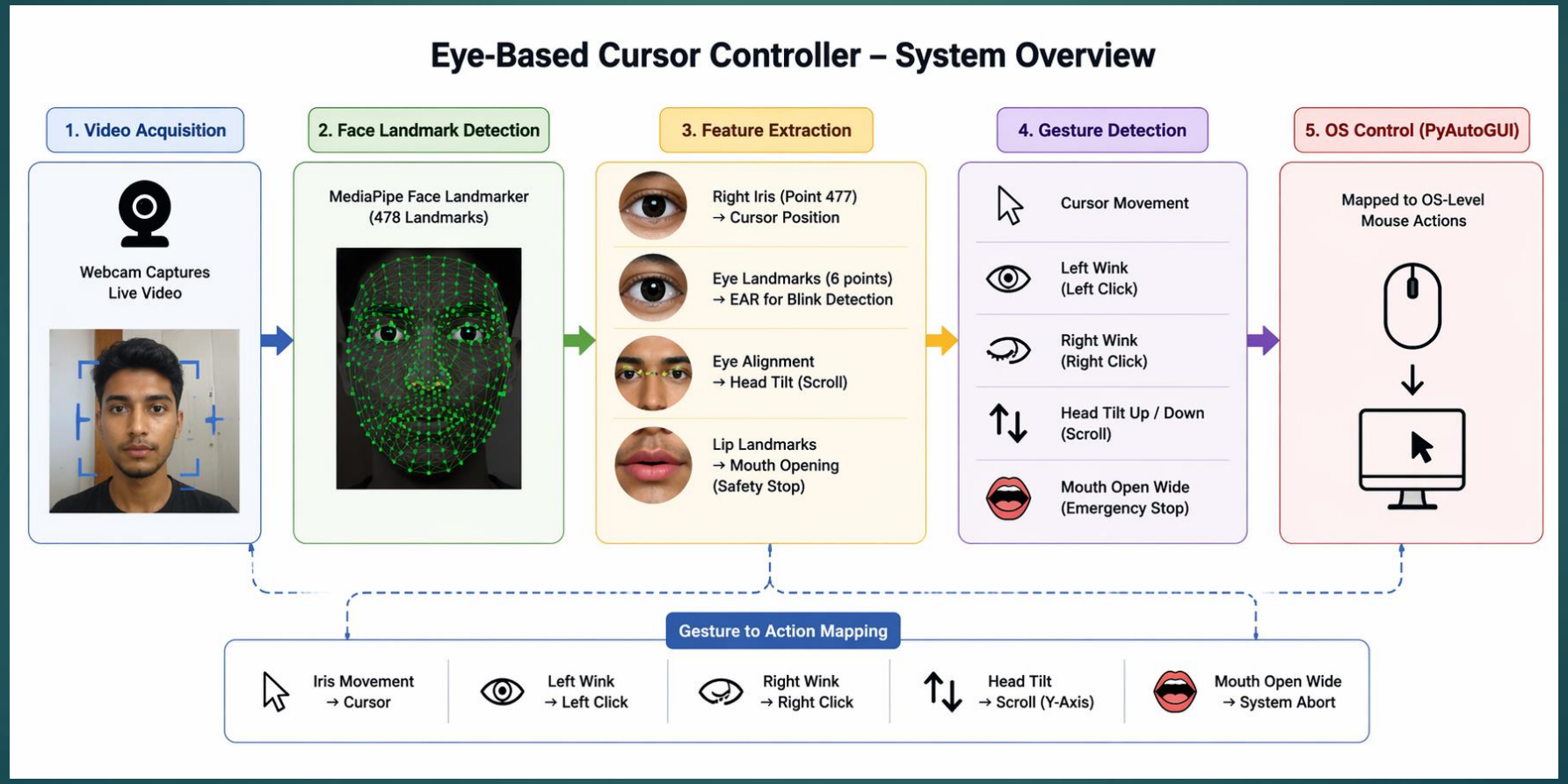
Scrolling

- Difference in eye heights
- Head tilt → vertical scrolling

Safety Mechanism

- Mouth opening detected
- Immediate program termination

System Overview



Design Choices & Key Decisions

Why EAR for Click Detection?

- Scale-invariant (independent of distance)
- More robust than pixel-based thresholds

Why Debounce Mechanism?

- Prevents false clicks from natural blinks
- Improves reliability

Why Iris Tracking for Cursor?

- More precise than face center tracking
- Smooth cursor movement

Why Threshold-Based Approach?

- Lightweight and real-time
- No training or calibration required

Libraries Used

- **OpenCV**

  Used for video capture and Frame Capture
- **MediaPipe**

  Used for its robust and fast Face Landmark Detection
- **PyAutoGUi**

  Allows easy integration with OS-level mouse control

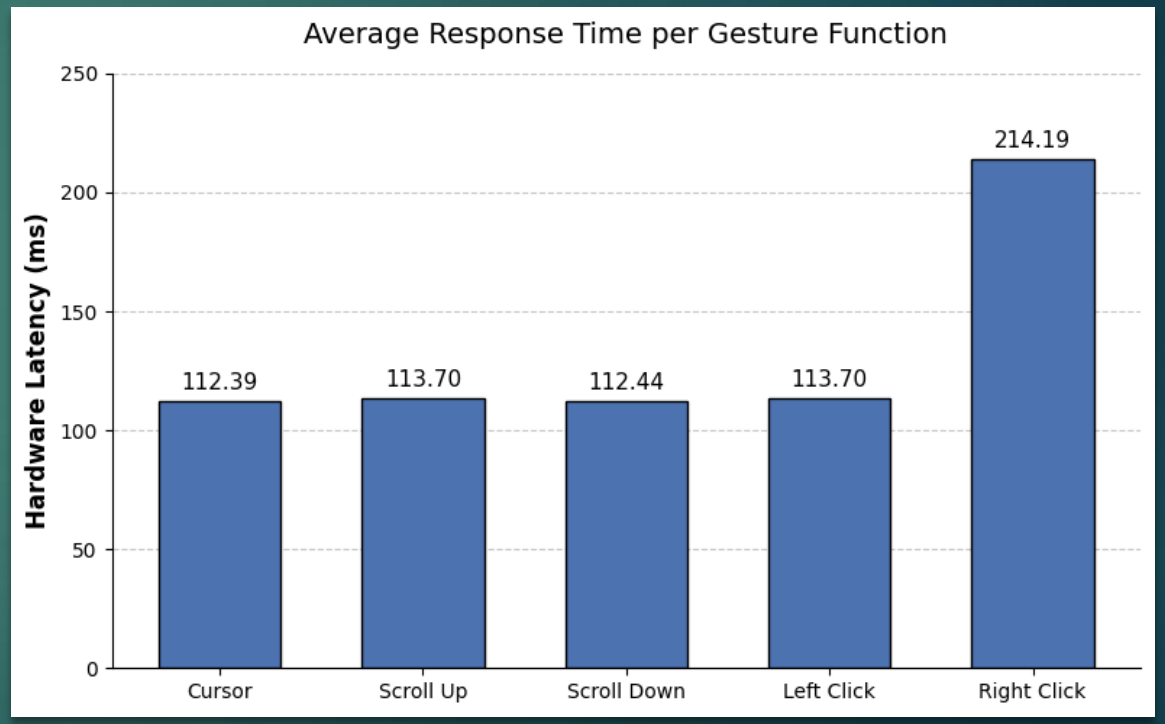
Experiments & Results

Experimental Setup

- Tested on standard laptop with webcam
- Real-time usage scenarios (cursor, click, scroll)
- Built-in telemetry for accuracy & latency

Key Results

- Average latency ≈ 125 - 140 ms
- Reliable cursor tracking in real time
- Accurate click detection using EAR + debounce
- Stable scrolling with head tilt



Analysis & Observations

Observed Issues

- Sensitive to poor lighting conditions
- Fixed thresholds may not generalize to all users
- Slight delay in rapid consecutive actions

Future Work

Future Work

- Adaptive thresholds for different users
- ML-based gesture classification (improve robustness)
- Better performance in low-light conditions
- Multi-face / multi-user support